

Turbo Equalization Based on Single Carrier QPSK Modulation for Underwater Acoustic Channel

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ABSTRACT

In this paper, the design of a Turbo equalizer is described. The equalizer is used as a part of the receiver when the QPSK modulation is used in underwater acoustic communications. The purpose of the equalizer is to remove the inter-symbol interference which is caused by the multipath propagation of the underwater acoustic signals. An laboratory simulations was conducted based on Proakis's B channel model. To verify the validity of the equalizer, a lake experiment is then carried out in June, 2015. During the experiment, QPSK signals were transmitted at the rate of 6000sps, and the range of the communication is from 700 meters to 2200 meters. The results of the experiment shows that the equalizer can effectively remove the ISI when the multipath propagation is not severe.

General Terms

Under water acoustic communications, QPSK, ISI

Keywords

Turbo equalizer, Multipath propagation, Sound speed profile

1. INTRODUCTION

Since the early 1980s, the underwater acoustic communications technology has progressed significantly [1]. Compared with terrestrial radio channel, underwater acoustic channel, characterized by limited transmitting bandwidth, severe multipath effect, and great dependence on underwater bottom structure and surface wind wave is a statistical space-time slow variation channel, making it a very difficult medium for communication [2]. Recently, Under Water Acoustic Sensor Networks (UW-ASNs) have been deployed to carry out collaborative monitoring tasks including oceanographic data collection, disaster prevention, and navigation [3]. One of the detrimental influence of the underwater acoustic channel is inter-symbol interference which is caused by the multipath effect of the propagation. Usually, an equalizer is required to remove the ISI. In this paper, an adaptive decision feedback equalizer based on Turbo code is described, the equalizer which is used in a MPSK modulation scheme includes a linear equalizer followed by a Turbo decoder, the output of the decoder is a soft decision value which will be feedbacked to the input of the linear equalizer, and after several iterations, the Turbo decoder outputs

the decision results. On the head of the transmitting frame, a small fraction of the signal is used as training sequence to estimate the channel. The LMS algorithm is used to update the feedforward and the feedback taps of the filter. To avoid the direct matrix inversion computation, the steep descent method is used to get the converge value of every tap. A simulation and a lake experiment were carried out to test the validity of the equalizer, the result proved that the equalizer can remove the ISI in moderate multipath interference.

The rest of the paper is organized as follows: Section 2 presents the principle of the equalizer. Section 3 describes the simulation results. Section 4 shows the performance of the equalizer during the lake experiment, and analyses the performance of the system. Finally, section 5 draws a conclusion.

2. PRINCIPLE OF THE SYSTEM

When the transmitting signals pass the underwater acoustic channel, the receive symbol is as following[4]

$$y_k = I_k + \sum_{\substack{n=0 \\ n \neq k}}^{\infty} I_n x_{k-n} + v_k$$

The term I_k represents the desired information symbol at the k th sampling instant, the term

$$\sum_{\substack{n=0 \\ n \neq k}}^{\infty} I_n x_{k-n}$$

represents the ISI, and V_k is noise term. To remove the ISI, we need a filter at receiving end, usually a decision feedback equalizer is used, which is presented as figure 1.

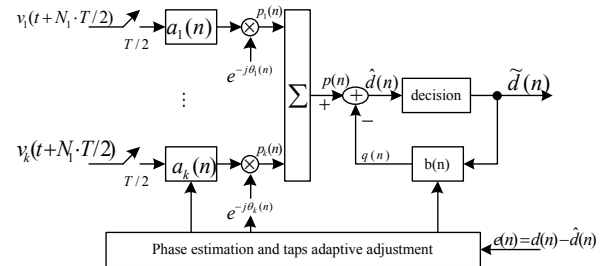


Fig.1. The diagram of the conventional DFE structure.

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The DFE performance degrades rapidly when the decision errors increase. To improve the performance of the equalizer, we deploy a Turbo decoder at the decision end. The received baseband signals firstly input one linear filter, the filter adjust the taps according to LMS algorithm and output the estimated symbol to the Turbo decoder. The decoder outputs the log-likely rate (LLR), and then transform it into the soft decision of the symbol, and feedback to the input of the filter. The iteration will be taken several times, then the decoder make symbol decision. The structure of the Turbo equalizer is shown in figure2.

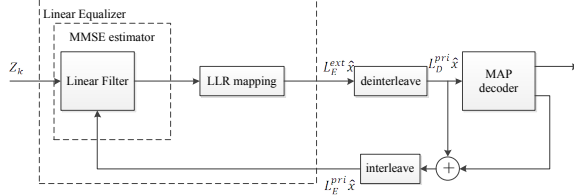


Fig.2. The structure of Turbo equalizer.

3. SIMULATION IN LABORATORY

To verify the validity of the equalizer, a simulation was taken in the laboratory. A Proakis' B type channel model was chosen as the channel impulse response, and a AWGN was assumed as the ambient noise factor. QPSK modulation was used to transmit the symbol at a speed of 6000 sps. Plot of error probability as a function of SNR/bit for different iterations are shown in figure 3, and we found that when the number of the iterations was more than five, there was no improvement of performance.

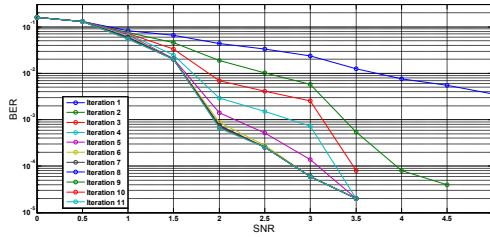


Fig.3. Simulation result of Proakis' B channel model.

4. LAKE EXPERIMENT RESULT

A lake experiment was carried out at Qiandao Lake, Zhejiang province in late of June, 2015. A acoustic modem was used to transmit QPSK signal at different distance through the boat, and another modem received the signal at fixed location. Because of the sunshine of the summer, there was a severe temperature variety depending on the depth of the water.

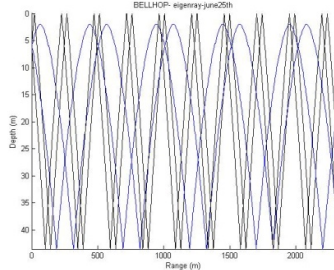


Fig.4. Track of the underwater acoustic signals.

A sound speed profile was measured before the communication experiment, and track of the sound propagation was simulated base on Bellhop model. Figure 4 shows the track of the signals.

According to the figures, there are no direct arrivals of the sound signal when the propagation range is more than 300 meters. The real communication range of the experiment was from 700m to 2200m, and we noted that when the range was more than 1300m, we could not get a reliable communication results. Figure 5 and figure 6 show the output constellation results of the equalizer when the communication distances were 750m and 900m.

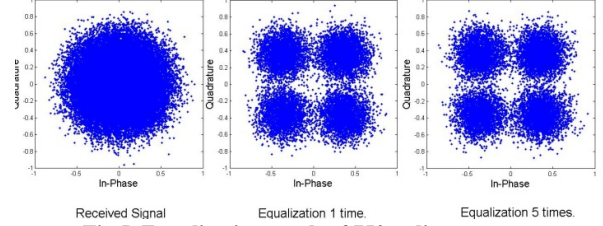


Fig.5. Equalization result of 750m distance.

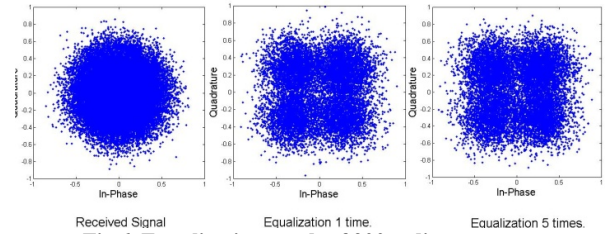


Fig.6. Equalization result of 900m distance.

5. CONCLUSION

We designed a Turbo equalizer to remove the ISI effect when communicating in the underwater acoustic channel. Compared to the conventional DFE, The Turboequalizer can get better error performance. Usually a space diversity is used to alleviate the deep fading of the signal, and an appropriate iteration number should be chosen to improve the performance of the Turbo equalizer without increase computation complexity.

6. ACKNOWLEDGEMENT

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